

1. Systems and Subsystems Required

To support a national-scale real-time payment platform handling **50,000+ TPS and millions of users**, the system must be composed of multiple distributed subsystems.

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Core Systems

1. User Management System

- User onboarding
- KYC verification
- Bank account linking
- Profile management

2. Authentication & Authorization System

- Identity verification
- Multi-factor authentication
- Token management

3. Payment Processing System

- Transaction validation
- Payment routing
- Real-time processing

4. Bank Integration System

- APIs for bank communication
- Interbank transaction routing
- Bank settlement coordination

5. Settlement & Reconciliation System

- Interbank settlement
- Daily reconciliation
- Transaction ledger management

6. Fraud Detection System

- Real-time fraud analysis
- Risk scoring
- Suspicious transaction monitoring

7. Notification System

- Transaction confirmations
- SMS/email/push notifications

8. Merchant Services

- Merchant onboarding
- QR code payments
- Merchant dashboards

9. Monitoring & Logging System

- System health monitoring
 - Transaction tracking
 - Alerting
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2. Information Exchanged Between Systems

Different systems exchange data at very high frequency.

User Data

- User identity
- Bank account details
- KYC verification data

Transaction Data

- Sender account
- Receiver account
- Transaction amount
- Transaction status

Banking Data

- Bank authorization responses
- Settlement records
- Reconciliation reports

Operational Data

- System logs
- Fraud alerts
- Monitoring metrics

Requirements

Frequency

- Thousands of requests per second

Speed

- Response time < 2 seconds

Volume

- Millions of transactions daily

Consistency

- Strong consistency for financial transactions
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3. Major Potential Failure Points

Technical Failures

- Database outages
- Server crashes
- Network latency

Integration Failures

- Bank API downtime
- API timeout errors
- Data format mismatches

External Dependency Failures

- Payment gateway failures
- Cloud provider outages
- DNS or network disruptions

Security Failures

- Fraud attacks
 - Unauthorized access
 - DDoS attacks
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4. Mitigation Strategies

To reduce failure risks, the architecture should include several mitigation strategies.

Redundancy

- Multiple data centers
- Replicated databases

Circuit Breakers

- Prevent cascading failures
- Temporarily block failing services

Load Balancing

- Distribute traffic across servers

Auto Scaling

- Scale infrastructure during high demand

Data Replication

- Maintain multiple copies of transaction data

Fraud Monitoring

- AI-based fraud detection
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5. Contingency Plans

When failures occur, the system must still ensure financial integrity.

Manual Reconciliation

- Compare transaction records
- Resolve mismatches

Deferred Settlements

- Temporarily queue transactions
- Process them later when systems recover

User Communication

- Inform users about delays
- Provide transaction status updates

Bank Communication

- Notify partner banks
 - Synchronize transaction states
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6. System Behavior Under Different Loads

Normal Daily Load

- Standard infrastructure capacity
- Regular monitoring
- Balanced system utilization

Peak National Events

During festivals or salary days, transactions may spike **10x**.

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System behavior should include:

- Automatic scaling of servers
 - Increased database capacity
 - Queue-based transaction buffering
 - Traffic throttling if necessary
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7. Architectural Trade-offs

Consistency vs Availability

Financial transactions require **strong consistency**, even if availability slightly reduces.

Cost vs Reliability

Higher reliability requires:

- Multi-region deployments
- Backup systems
- Disaster recovery infrastructure

This increases cost but ensures system stability.

Real-time vs Eventual Processing

Core transactions must be **real-time**, but analytics and reporting can use **eventual consistency**.

Security vs User Experience

Strong security measures like MFA increase safety but may slightly affect user convenience.

8. Global Implementations & Additional Features

Several countries have implemented similar systems.

Examples

UPI (India)

- Instant mobile payments
- QR-based transactions

Faster Payments (UK)

- Real-time bank transfers

FedNow (USA)

- Instant payments infrastructure

Additional Features Recommended

AI Fraud Detection

- Detect suspicious patterns instantly

Cross-Border Payments

- Enable international transactions

Smart Routing

- Automatically select fastest bank route

Advanced Analytics

- Real-time financial insights

Business Impact & ROI

- Increased digital payment adoption
- Reduced cash dependency
- Faster business settlements
- Improved financial transparency
- Higher economic efficiency